Complementizer Variation in Guadeloupean Creole:

An Empirical and Experimental Approach

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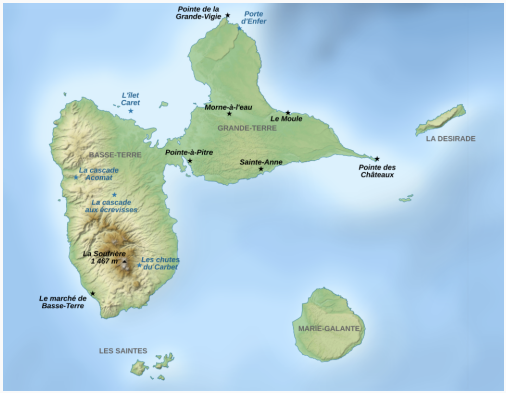
Overview

1. Introduction
2. Experimental Methodologies
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5. Interview Results
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Introduction

Guadeloupean Creole





Guadeloupean Creole is a French-lexified Antillean Creole language spoken in the French department of Guadeloupe (Colot & Ludwig, 2013; Wylie, 1995).

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- We conducted fieldwork in Guadeloupe to analyze this variation:
  - Acceptability judgments on variants through an **forced-choice task experiment.**
  - Natural variation rates gathered via **sociolinguistic interviews.**

Background

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- Would youth educated in both GC and French be **less likely to use the *kè* variant?**

Complementizer variation

(1) a. The complementizer variant ***kè*** in GC [1]

Marin déklaré ***kè*** *yo gannyé chanpyona-la*

Marine declare.PFV COMP 3PL win.PFV championship-DET

'Marine declared that they won the championship.'

[1] Examples taken from fieldwork

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'Marine declared that they won the championship.'

b. The complementizer variant **Ø**

Marin déklaré ***Ø*** *yo gannyé chanpyona-la*

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'Marine declared they won the championship.'

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French & GC Complementation

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(2) French (Bonan, 2013)

Jean savait qu'il allait mourir.

Jean know.PST COMP-3SG go.PST die

'Jean knew that he was going to die.'

(3) GC (Bonan, 2013)

Jan té sav Ø i té kay mò.

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- Recent observations, however, suggest that the *kè* variant of the complementizer can be **acceptable to GC speakers** (Tramutoli, 2021).

Research Questions

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- **RQ 2:** In their choice of complementizer variant, are adult speakers **consistent** in their production between experiments and natural speech?

Experimental Methodologies

Data Collection: Experiment

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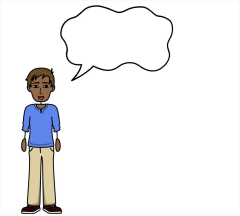
- 9 **youth** speakers, age 11–13
  - Two schools: one monolingual French, the other also offering oral GC classes.
- 11 **adult** speakers, age 18–64
  - Recruited through word of mouth and flyers.
  - Included educators, administrative staff, janitors, and others.

Experiment: Trial Procedure

1. An audio plays of a complex sentence with a quotative, with one gender introducing the quote and the opposite speaking it.

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Step 1: Quotative

Filip di: "An enmé lapli."

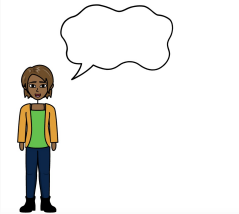
"Filip said: ‘he likes rain.’"

Experiment: Trial Procedure

2. The same sentence plays aloud as a statement rather than a quotative, either with or without overt *kè*.

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Step 2a: Without *kè*

Filip di i enmé lapli.

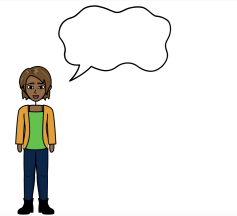
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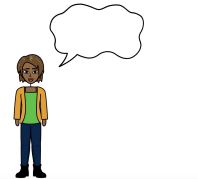
Step 2b: With *kè*

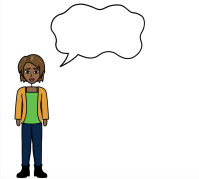
Filip di **kè** i enmé lapli.

Filip said that he likes rain.

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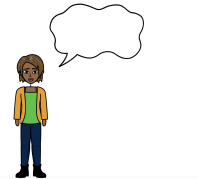
4. The participant makes a binary choice between the two variants.

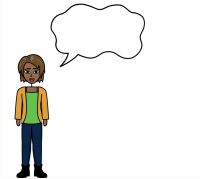




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The entire per-trial procedure:

Version	Step	Sentence	Translation
Direct speech	Step 1	Filip di : “An enmé lapli”	Filip said: “I like rain”
Variant: zero	Step 2a	Filip di i enmé lapli.	Filip said he likes rain.
Variant: kè	Step 2b	Filip di **kè** i enmé lapli.	Filip said that he likes rain.

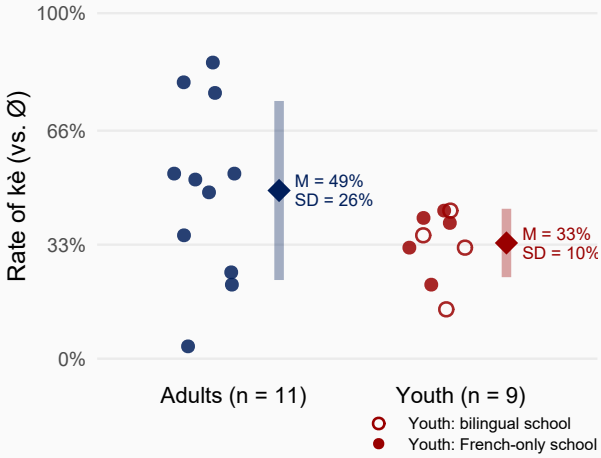
Verbs in Experimental Stimuli

Verb (GC)	English	Total instances
di	to say	5
déklaré	to declare	3
kriyé	to shout	3
rimaké	to point out	3
ajouté	to add	2
anonsé	to announce	1
avwé	to admit	1
bougonné	to grumble	1
chichoté	to whisper	1
èksplikè	to explain	1
hélé-anmwé	to cry out	1
konfésé	to confess	1
noté	to note	1
pansé	to think	1
pléré	to cry	1
sonjé	to remember	1
vansé	to put forward	1
Total		*28*

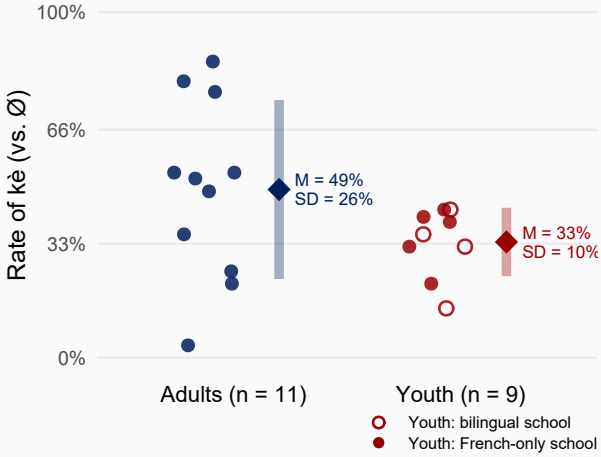
Table 1: Verbs preceding COMP variant in the stimuli, sorted by frequency across sentences.

Experimental Results

Experiment: Overall Results

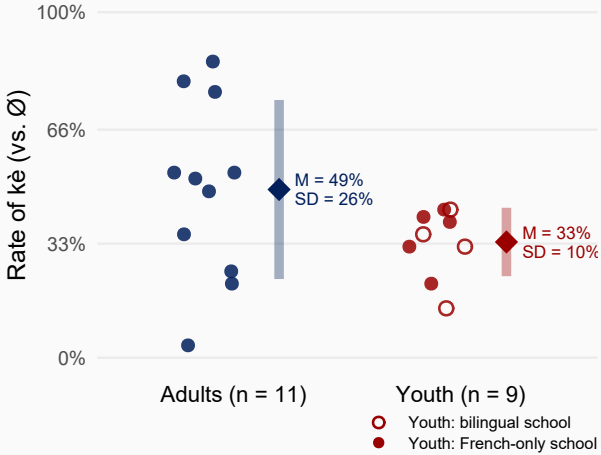


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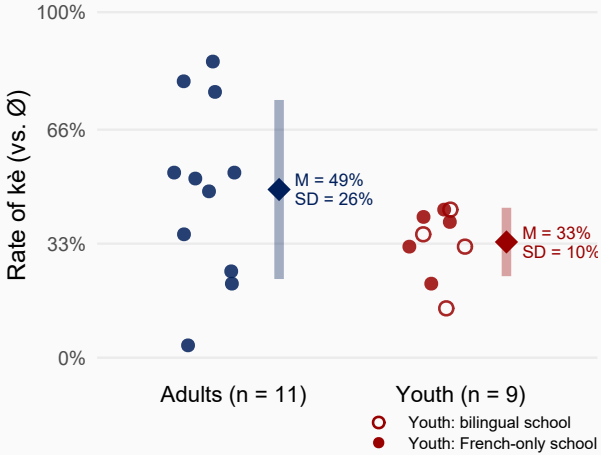
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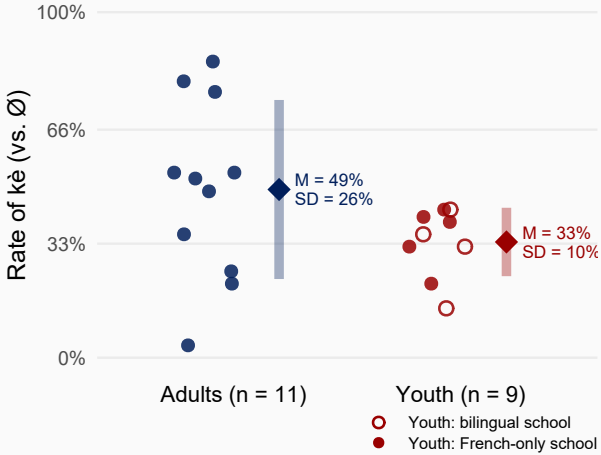
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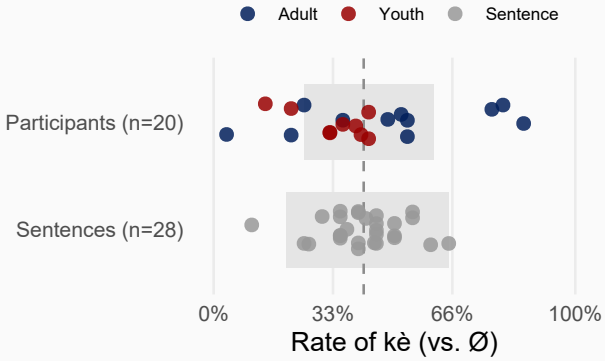
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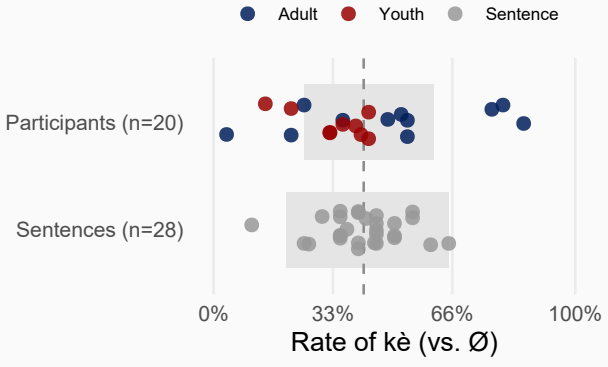
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  - Adults **vary widely** (4–86%)
  - Youth are **more consistent** (14–43%)

Sources of Variation: Speakers vs. Sentences



Grey band: 95% range expected if all shared the same rate.
Dashed line: overall rate.

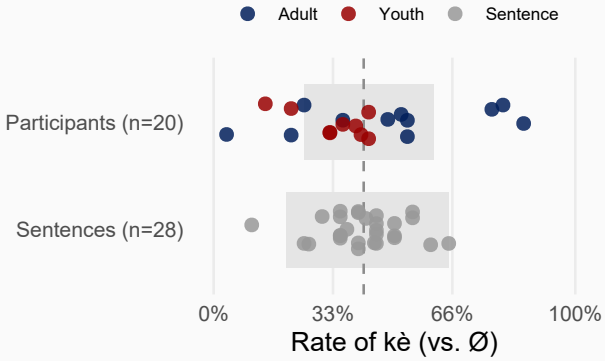
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- **Sentences** vary no more than chance (SD 11 vs. 11 points, $p = .44$).

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- **No,** youth did not differ significantly based on schooling type ($p = .73$).

Interview Protocol

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NLP Analysis of Transcriptions

- Adult interviews in GC were **analyzed for *kè* vs. Ø.**

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A4	11:49
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A7	10:35
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A10	13:15
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Table 2: Adults selected for interview analysis.

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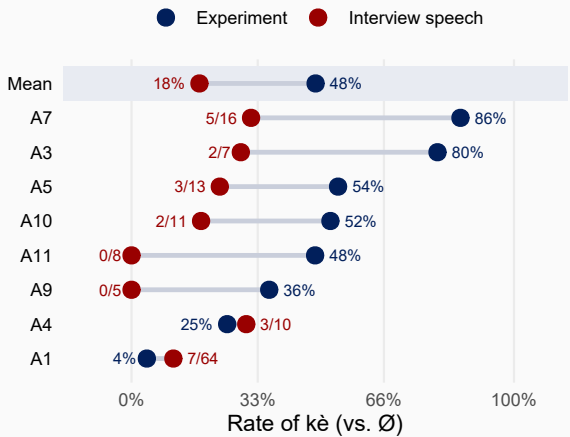
- 8 interviews were analyzed with a custom Python pipeline built on NLTK (Bird et al., 2009).

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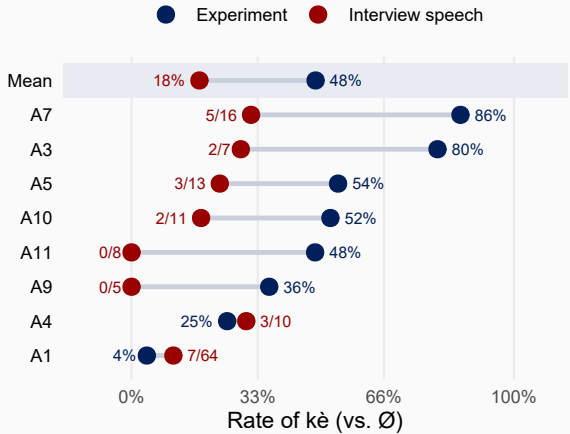
Interview Results

Comparative Results: Experiment vs. Speech



Interview labels: target verbs with kè / all target verbs.

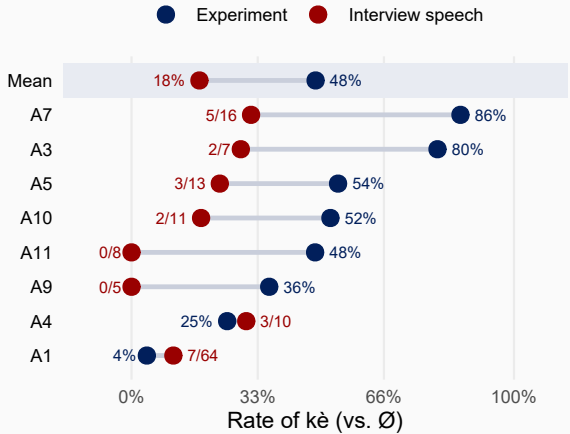
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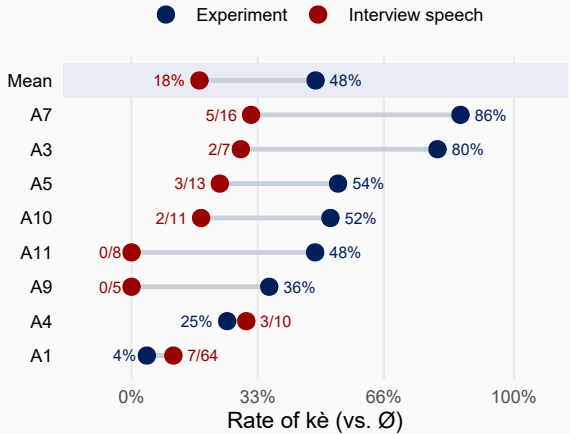
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Experimental choices and production in interview are **not significantly correlated** ($\rho = .51, p = .19, n = 8$).

RQ 2 Revisited

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No: adults chose *kè* **far more often in the experiment** than they used it in speech (48% vs. 18%, $p = .04$).

Discussion

Bilingual Education in Guadeloupe

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Youth could be **adopting low rates of naturalistic *kè* production seen in adults**:

- Youth experimental rates (14–43%) sit much closer to adult interview rates (0–31%) than to adult experimental rates (4–86%).

Conclusion

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   - Adults **vary widely** (4–86%); youth do not (14–43%), regardless of school type.

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   - Adults **vary widely** (4–86%); youth do not (14–43%), regardless of school type.
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   - Though based on a small sample, this suggests *kè* variation may be transmitted more through everyday speech than through schooling.

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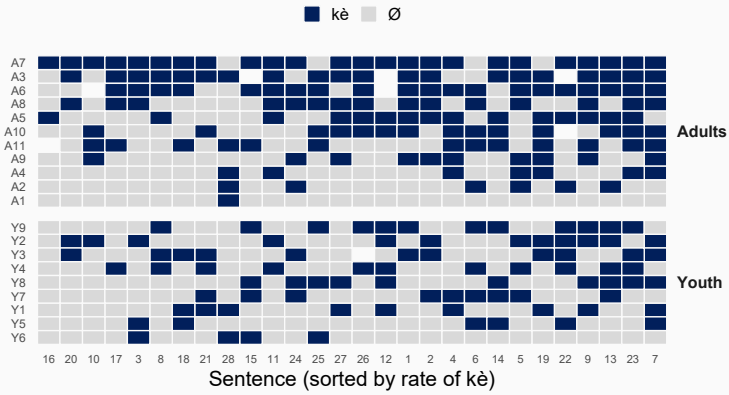


Penn Language Contact and Cognition Lab
Penn Language Variation and Cognition Lab

STAY IN TOUCH :)

Appendix

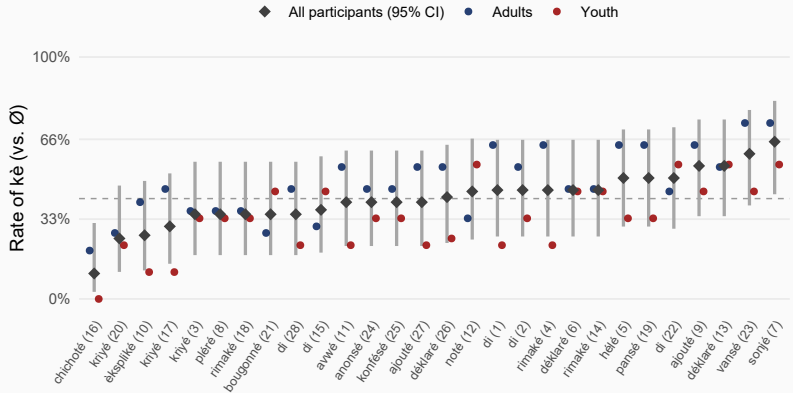
Experiment: Every Response



Horizontal bands: **participants** differ consistently.

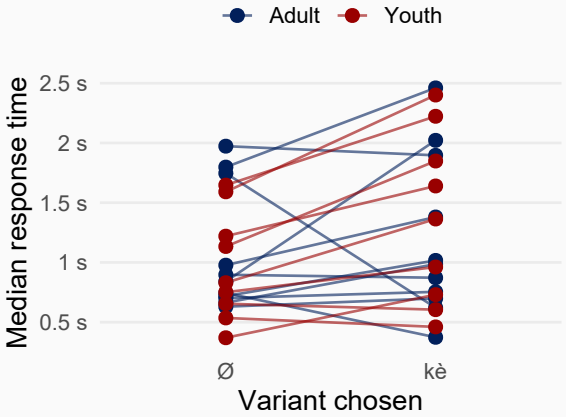
Few solid vertical bands: **sentences** matter far less.

Experimental Results: Stimuli Sentences



Only *chichoté* "whisper" (11%) and *sonjé* "remember" (65%) fall clearly outside the overall mean (dashed line, 41.5%).

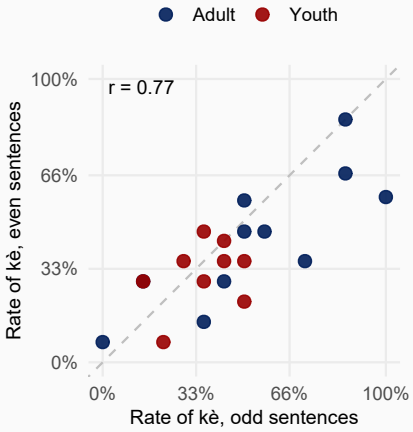
Response Time



14 of 20 speakers were slower when choosing kè.

- In the experiment, choosing *kè* took about **19% longer** ($p = .02$); 14 of 20 participants were slower.
- This suggests that *kè* might be a more **deliberate, metalinguistic** choice.

Experiment: Individual Consistency



- Each participant's *kè* rate on odd- vs. even-numbered sentences.
- The splits agree strongly ($r = .77, p < .001$): individual rates are **stable preferences**.
- Driven by adults ($r = .81$); youth are less consistent ($r = .35$) per-participant within their narrower range of responses.

What Predicts *kè* at the Sentence Level?

Sentence property	Rate of *Kè* (no. of sentences)						χ^2(df)	*p*
Verb type	Cognition	57% (2)	Utterance	40% (26)			4.80 (1)	**.028**
Matrix TMA	Marked	58% (2)	None	40% (26)			5.47 (1)	**.019**
Embedded subject	Pronoun	43% (17)	NP	39% (9)	Existential	42% (2)	0.69 (2)	.709
Adverb before *kè*	Yes	40% (2)	No	42% (26)			0.06 (1)	.809
Causative *fè*	Yes	45% (2)	No	41% (26)			0.19 (1)	.664
Second *kè* in sentence	Yes	35% (1)	No	42% (27)			0.42 (1)	.519
Group × verb type	Adults and youth respond alike to verb type						0.57 (1)	.450

Table 3: Each property tested with a likelihood-ratio test (mixed-effects logistic models, random intercepts for participant and item).

- However, verb type and TMA each rest on only **2 sentences.**
  - Both of these include *sonjé* (7) “remember”.

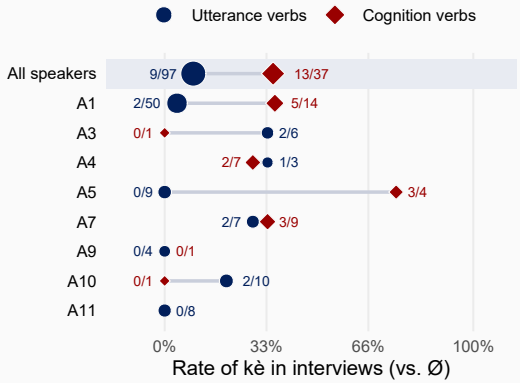
Type-Token Ratio Corpus Analysis

Participant	Tokens	Types	TTR	MATTR
A1	2508	452	0.180	0.555
A3	582	173	0.297	0.521
A4	858	203	0.237	0.500
A5	1133	259	0.229	0.565
A7	905	234	0.259	0.544
A9	988	245	0.248	0.534
A10	1410	296	0.210	0.524
A11	994	279	0.281	0.557
Mean	*1172*	*268*	*0.243*	*0.538*

Table 4: Adult interview transcripts. TTR = type–token ratio; MATTR = moving-average TTR.

- **TTR drops as transcripts get longer**: A1 has the longest interview and the lowest TTR (0.180).
- **MATTR is stable** across speakers (0.50–0.57), so it is the fair way to compare lexical diversity across interviews of different lengths (Covington & McFall, 2010).

Naturalistic Verb Type Analysis



Labels: tokens with kè / total tokens.
Dot size = number of tokens.

- Pooled across speakers, *kè* follows **cognition verbs** far more often than utterance verbs (35% vs. 9%, $p < .001$), in line with Tramutoli (2021).
- Mostly driven by *pansé* ‘think’ (12/26)
  - By contrast, *sonjé* ‘remember’ rarely takes *kè* in speech (1/11), despite drawing the highest *kè* rate in the experiment (65%).
- Per-speaker counts are small (often 1–4 tokens); only A1 has many of both types (5/14 vs. 2/50).